COMP/SOFT4902/4912 Graduation Design Project	FEYZİYE SCHOOLS FOUNDATION IŞIK UNIVERSITY

Tumor Segmentation Using Deep Learning

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REQUIREMENTS ANALYSIS DOCUMENT

1. Introduction

1.1. Purpose of the System

The purpose of this project is to develop a desktop-based software system that assists in the analysis of mammography images by identifying suspicious regions and providing a prediction about whether the detected finding is benign or malignant. The system is intended as a decision-support prototype rather than a replacement for medical experts. Its key role is to help users inspect breast images more efficiently by combining image preprocessing, lesion localization/segmentation, classification, and visual explanation in a single workflow.
This system is being developed because breast image interpretation can be time-consuming and error-prone, especially when findings are subtle. By offering a structured and automated analysis pipeline, the project aims to improve consistency, reduce manual effort, and demonstrate how deep learning can be used in a practical healthcare-related application.

1.2. Scope of the System

The scope of this project includes the design and implementation of a desktop application for breast image analysis. The system will accept breast imaging input, apply preprocessing operations, detect or segment suspicious regions, classify the detected finding as benign or malignant, and present the result together with an explainability output to show which areas influenced the model's decision. The project also includes dataset preparation, model training, testing, evaluation, and documentation of the system.

The project is limited to a prototype environment and publicly available datasets. It does not aim to become a certified clinical product, integrate directly with hospital information systems, or provide final medical diagnoses. Real-time deployment in hospitals, doctor/patient account management, cloud infrastructure, mobile support, and large-scale clinical validation are outside the scope of this project. The system is also not intended to handle every possible medical imaging modality or every cancer type; the current focus is mammography images as defined in the proposal.

1.3. Objectives and Success Criteria of the Project

The main objective of the project is to build an end-to-end prototype that can process mammography images and produce meaningful outputs for both localization and classification. A second objective is to make the system interpretable by adding a visual explanation component instead of presenting the model as a black box. Another objective is to gain practical experience in medical image preprocessing, segmentation architectures such as U-Net, classification methods, and evaluation in a high-stakes domain.
The project will be considered successful if the following conditions are met:

- A working desktop application is delivered.
- The application can accept breast imaging input and complete the full pipeline from preprocessing to output generation.
- The system can localize or segment suspicious regions and provide a benign/malignant classification result.
- The result screen includes an explainability visualization that helps the user understand the model output.
- The model is evaluated on a test set, and its performance is reported using appropriate metrics for segmentation and classification.
- The system runs reliably enough to be demonstrated to the supervisor and jury without major functional errors.

1.4. Definitions, Acronyms, and Abbreviations

1.5. AI: Artificial Intelligence

1.6. CNN: Convolutional Neural Network

1.7. DICOM: Digital Imaging and Communications in Medicine, a standard format used for medical images

1.8. Dice Score: A metric used to measure similarity between predicted and ground-truth segmentation masks

1.9. Grad-CAM: Gradient-weighted Class Activation Mapping, a method used to visualize which image regions influenced a model prediction

1.10. GUI: Graphical User Interface

1.11. IoU: Intersection over Union, a metric used to evaluate segmentation overlap

1.12. RAD: Requirements Analysis Document

1.13. ROI: Region of Interest

1.14. U-Net: A convolutional neural network architecture widely used for medical image segmentation

1.15. Benign: A non-cancerous abnormal finding

1.16. Malignant: A cancerous abnormal finding

1.17. Prototype: An initial functional version of a system built to demonstrate feasibility and core features

1.18. Overview

This RAD presents the requirements and analysis of the proposed breast tumor analysis system. It describes the purpose and scope of the project, the current way the problem is handled, the proposed system, and the main functional and non-functional requirements. It also includes system models such as use case diagrams, class diagrams, and user interface mock-ups.

The document is organized in a way that starts with the general project background and continues with more detailed system analysis. First, it introduces the purpose, scope, and objectives of the project. Then, it explains the current situation and the proposed solution. After that, it presents the system requirements and analysis models that will guide the design, implementation, testing, and final evaluation of the system.

2. Current System

Currently, breast image analysis is mostly done manually by radiologists. They examine the images, look for suspicious areas, and decide whether the finding might be benign or malignant. This method is widely used, but it depends a lot on human experience and can be time-consuming.
There are also some existing computer-based and research-based solutions. However, many of them only focus on one part of the problem. For example, some only classify the image, while others only

detect or segment suspicious regions. Many of these systems also do not clearly show why the model made a certain decision.
Because of this, the current way of dealing with the problem is still limited. It often requires a lot of manual effort, and existing tools are not always complete or easy to interpret. This project aims to improve that by combining segmentation, classification, and visual explanation in one desktop-based system.

3. Proposed System

3.1. Overview

The proposed system is a desktop-based medical imaging application designed to assist healthcare professionals in the detection and diagnosis of breast tumors. The system accepts mammography images as input and runs them through a three-stage AI pipeline: preprocessing, segmentation, and classification, followed by an explainability layer that visualizes the regions that influenced the model's decision.

The primary stakeholder is the medical professional (e.g., radiologist or clinician) who interacts with the desktop application directly. They upload scan images, trigger the analysis, and interpret the results. There are no administrative roles or multi-user management features within the scope of this project. The system operates fully locally on a personal computer with no network dependency at inference time.

3.2. Requirements

The system shall allow a medical professional to load a mammography image in a supported format (PNG, TIFF, or DICOM converted to a standard format). Once loaded, the system shall preprocess the image automatically to prepare it for analysis. The system shall then perform tumor segmentation, identifying and delineating suspicious regions within the image. Following segmentation, the system shall classify the detected region as either benign or malignant. Finally, the system shall provide an explainability output that visually indicates which regions of the image influenced the model's decision, allowing the medical professional to interpret the result.

3.2.1. Use Cases

UC1: Load Mammography Image: Priority: High, the medical professional launches the application and selects a mammography image file from their local file system. The system validates the file format and loads the image, displaying it in the application's image viewer panel.

UC2: Preprocess Image: Priority: High, After the image is loaded, the system automatically preprocesses it to prepare it as suitable input for the segmentation model. The user does not need to trigger this manually; it is initiated as part of the analysis pipeline.

UC3: Segment Tumor Region: Priority: High; the system runs the preprocessed image through the segmentation model, which identifies and delineates suspicious regions. The output is a segmentation mask overlaid on the original image, visually highlighting the detected area of interest.

UC4: Classify Detected Region: Priority: High, Using the segmented region as input, the system runs the classification model and produces a label, benign or malignant, along with a confidence score indicating the model's certainty.

UC5: Visualize Explainability Output: Priority: High, the system generates an explainability visualization and overlays it on the original image, highlighting the regions that most influenced the classification decision. This is displayed alongside the classification result to allow the medical professional to interpret the model's reasoning.

UC6: Export Results: Priority: Medium, the medical professional can export the analysis results including the annotated image, segmentation mask, classification label, confidence score, and explainability heatmap as a report file to their local system.

3.2.2. Scenarios

UC1: Load Mammography Image

Scenario 1: Successful Image Load: The medical professional opens the application and clicks the load button. They browse their file system and select a valid mammography image in PNG format. The system validates the file and displays it in the image viewer panel, ready for analysis.

Scenario 2: Unsupported File Format: The medical professional attempts to load a file in an unsupported format (e.g., a JPEG or a raw DICOM without prior conversion). The system detects the invalid format and displays an error message informing the user that the file is not supported, without crashing or proceeding to analysis.

UC3: Segment Tumor Region:

Scenario 1: Suspicious Region Detected: The system runs the preprocessed image through the segmentation model and identifies a suspicious region. A segmentation mask is overlaid on the image, visually delineating the detected area.

Scenario 2: No Suspicious Region Detected: The segmentation model finds no suspicious regions in the image. The system informs the user that no abnormality was detected and does not proceed to classification.

UC4: Classify Detected Region

Scenario 1 High Confidence Classification: The system classifies the segmented region as malignant with a high confidence score. The result is displayed alongside the explainability of visualization.

Scenario 2 Low Confidence Classification: The system produces a classification result but with a low confidence score. The system displays the result while flagging the low confidence to the user, suggesting further review.

UC5: Visualize Explainability Output:

Scenario 1 Successful Explainability Visualization: Following a successful classification, the system generates the explainability visualization and overlays it on the original image. The highlighted regions are displayed alongside the classification label and confidence score, allowing the medical professional to interpret the model's reasoning.

Scenario 2 Explainability Generation Fails: The system completes classification but fails to generate the explainability visualization. The system displays the classification result and confidence score as normal but notifies the user that the explainability output could not be generated, without crashing or affecting the classification result.

UC6: Export Results

Scenario 1 Successful Export: After the analysis pipeline completes, the medical professional clicks the export button. A file save dialog appears and they select a destination folder on their local system. The system compiles the results and saves the report file successfully, notifying the user that the export is complete.

3.3. Nonfunctional Requirements

Usability: The application shall provide a simple, straightforward desktop interface that allows a medical professional with no prior technical background in AI to complete a full analysis from image loading to result in interpretation without any training. Error messages shall be clear and actionable, not technical.

Reliability: The system shall handle invalid inputs, unsupported file formats, and model inference failures gracefully without crashing. The segmentation and classification results shall be reproducible, meaning the same image input shall always produce the same output.

Performance: The full pipeline of preprocessing, segmentation, classification, and explainability generation shall be completed within a reasonable time on a standard personal computer. The system shall not require a GPU to run, though GPU availability may improve processing speed.

Supportability: The codebase shall be modular, separating the UI, preprocessing, model inference, and explainability components, so that individual components can be updated or replaced without affecting the rest of the system.

Implementation: The system shall be implemented in Python. The desktop interface shall be built using a standard Python GUI framework. Deep learning components shall use PyTorch or TensorFlow.

Legal: All datasets used for training and evaluation shall be publicly available and used in compliance with their respective licenses. No patient-identifiable information shall be stored or processed by the application.

Other Constraints: The system directly addresses SDG 3 (Good Health and Well-Being) by supporting early breast cancer detection in clinical workflows. It has potential to reduce diagnostic workload in under-resourced healthcare settings where specialist availability is limited. The system is a decision-support tool and shall not be presented as a replacement for professional medical judgment.

3.4. System Models

3.4.1. Use Case Model

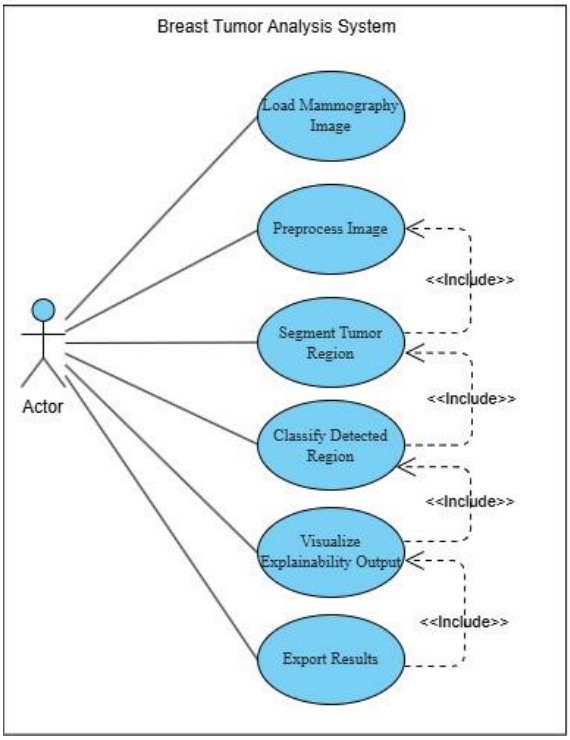


Figure 1 Use Case Diagram

3.4.2. Object Model

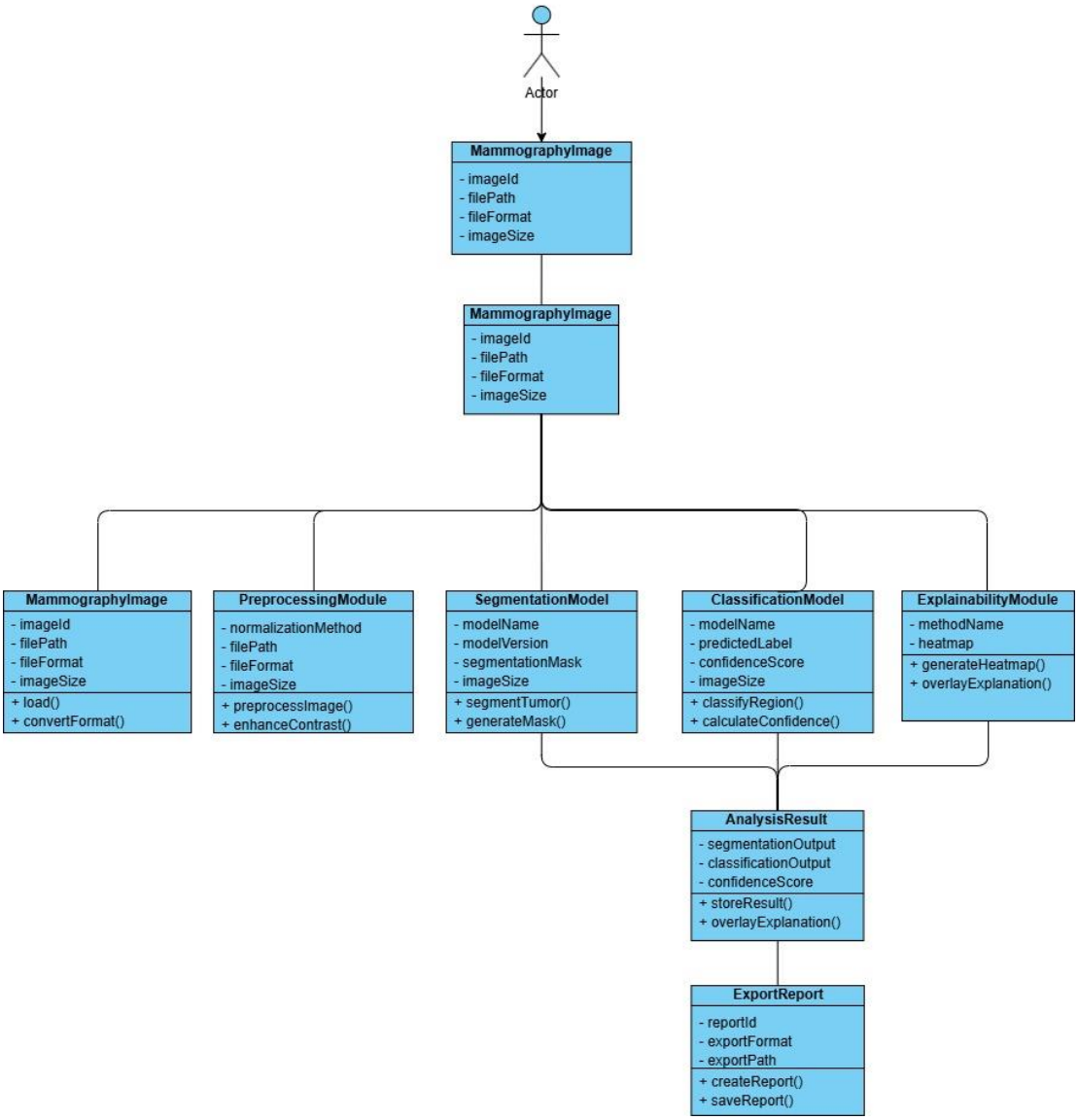
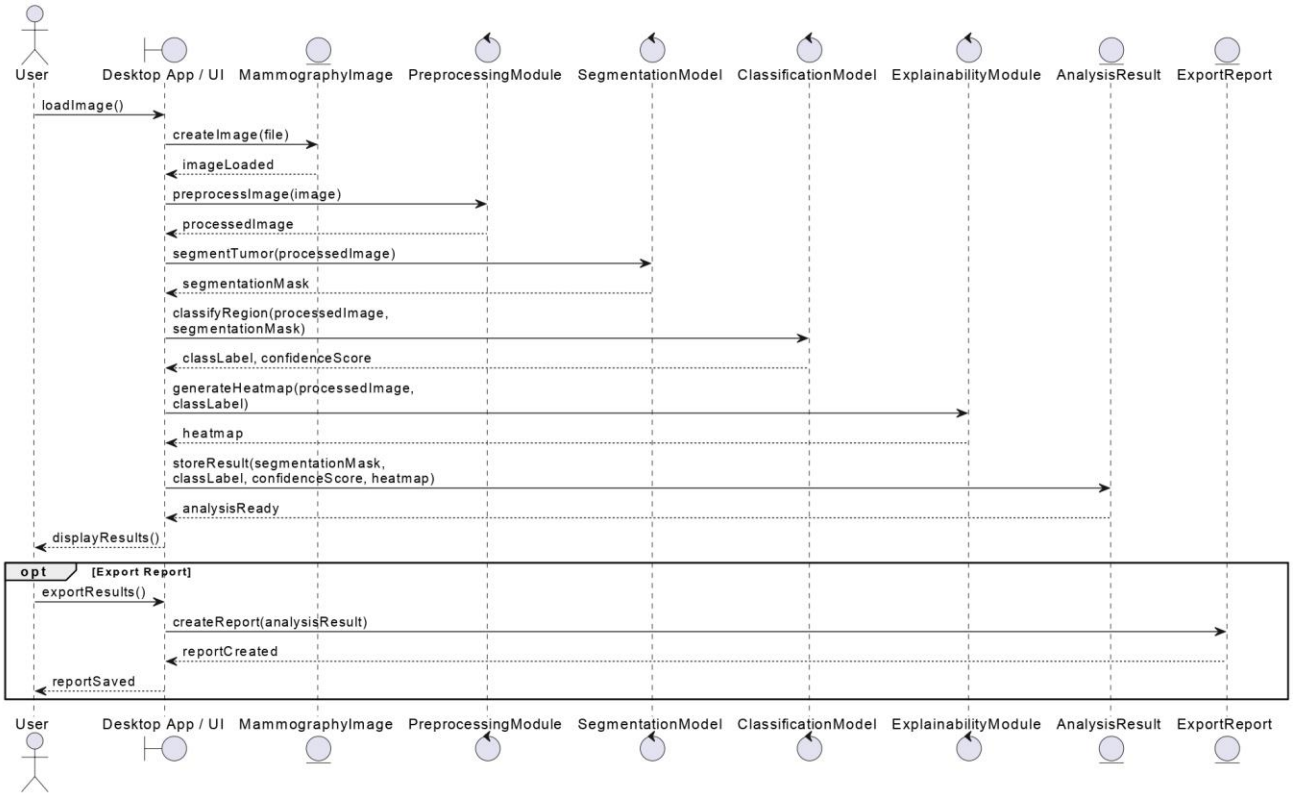


Figure 2 Class Diagram

3.4.3. Dynamic Model



3.4.4. User Interface—Navigational Paths and Screen Mock-ups

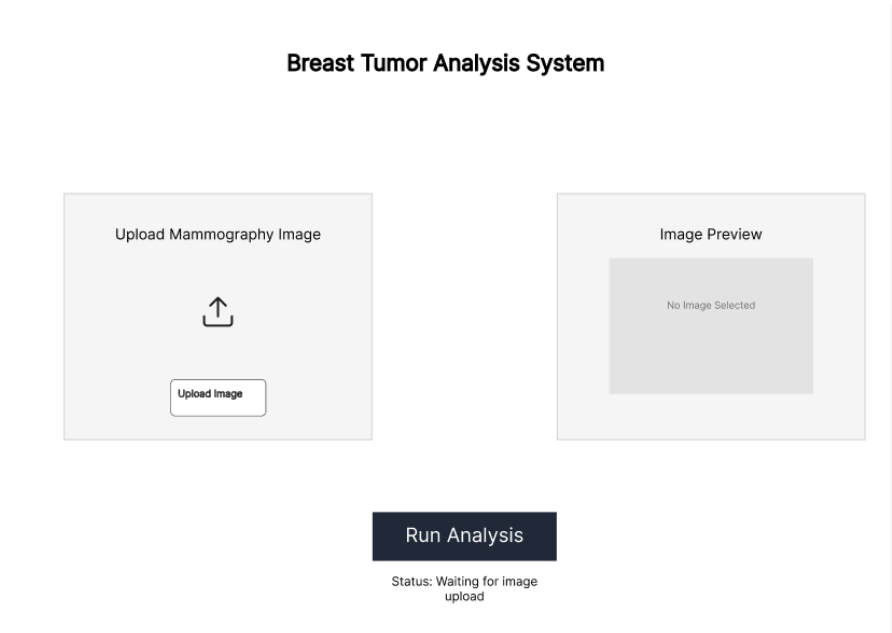


Figure 3 Main Screen of the desktop app

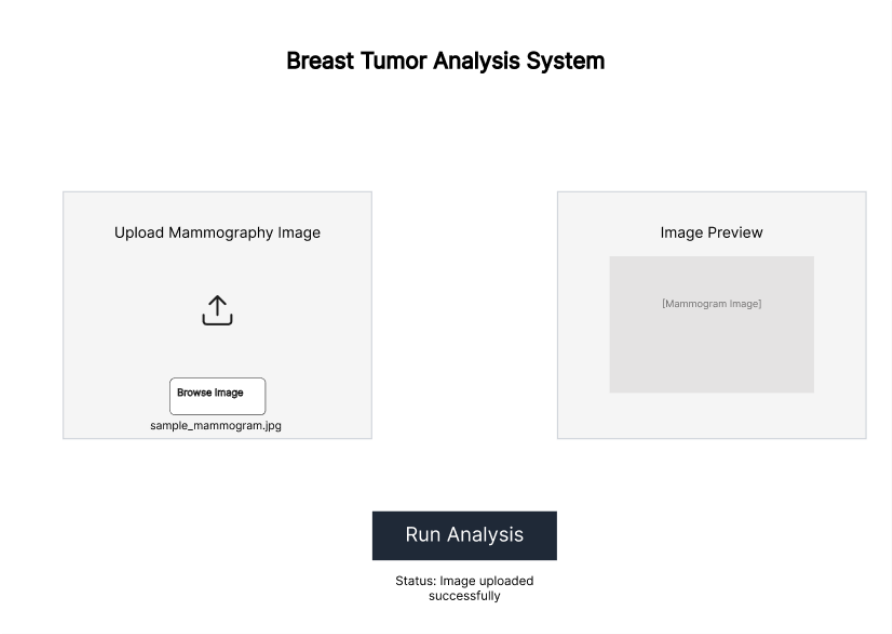


Figure 4 Desktop screen after an image uploaded

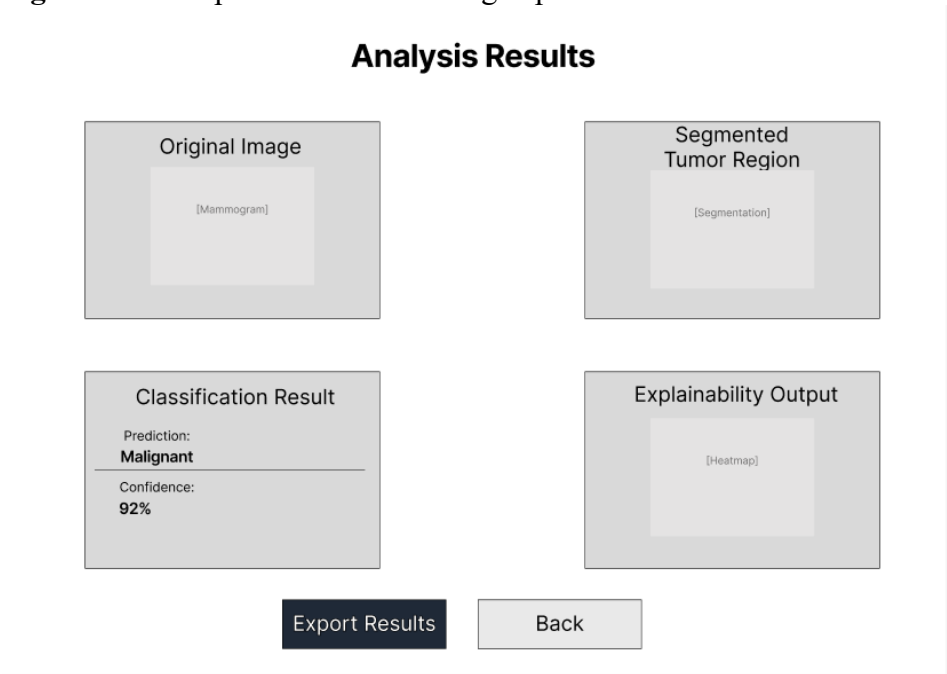


Figure 5 Results screen after analyzing mammography images

4. Other Analysis Elements

4.1.Risks and Alternatives

Explain possible risks that you might encounter during the project and list your alternative plans in case such risks do occur, with a template given in Table 2 below.

Table 1. Risks

Risks	Likelihood	Effect on the project	B Plan Summary
Desktop application	Medium	Medium, poor user experience	experience Optimize model size through techniques like

performance is too slow on a CPU-only machine			quantization or pruning, or a GPU as a minimum requirement
Segmentation model fails to achieve acceptable performance on the chosen dataset	Medium	High, unreliable segmentation breaks the entire pipeline	Simplify the task to detection/localization using bounding boxes instead of pixel-level segmentation
Scope creep due to unclear boundaries between segmentation and classification components	High	Medium, delays implementation	Strictly follow the modular architecture defined in the design phase and freeze requirements early
Explainability output is not meaningful or interpretable enough for medical context	Medium	Medium, weakens the clinical value of the system	Explore alternative XAI techniques or present the output as supplementary rather than a core feature

1.19. Project Plan

Table 2. List of work packages

WP#	Work package title	Members involved
WP1	Requirements Analysis and Planning	Dilara, Abdelrahman
WP2	System and Model Design	Dilara, Abdelrahman
WP3	Data Preparation and Preprocessing	Dilara, Abdelrahman
WP4	Segmentation and Classification Implementation	Dilara, Abdelrahman
WP5	Desktop Application Integration and Testing	Dilara, Abdelrahman
WP6	Documentation, Thesis Writing, and Presentation	Dilara, Abdelrahman

Table 3. Work Package 1

WP 1: *Requirements Analysis and Planning*			
Start date: 02*.03.2026* **End date:** *15.03.2026*			
Leader:	Dilara Sönmez	**Members involved:**	*Dilara Sönmez* *Abdelrahman Elmughrabi*
Objectives: The objective of this work package is to establish the foundation of the project by defining the system scope, identifying stakeholders, and documenting all functional and non-functional requirements. This includes analyzing the problem domain, identifying use cases, and			

assessing potential risks. The work package concludes with the preparation of the Requirements Analysis Document, which will guide all subsequent design and implementation phases.
Tasks: ***Task 1.1 Requirements for elicitation and system scope definition:*** *Identify the project boundaries, user needs, and major system functions.* ***Task 1.2 Functional and non-functional requirements definition:*** Document the system behavior, constraints, and quality requirements. ***Task 1.3 Risk analysis:*** Identify possible technical and project-related risks and prepare alternative plans. ***Task 1.4 RAD preparation:*** Compile the requirements of analysis results into the RAD document.
Deliverables ***D1.1:*** Requirements list ***D1.2:*** Risk analysis table ***D1.3:*** Requirements Analysis Document ***D1.4:*** *Diagrams*

Table 4. Work Package 2

WP 2: *System and Model Design*			
Start date: *16.03.2026* **End date:** *27.03.2026*			
Leader:	*Abdelrahman Elmughrabi*	**Members involved:**	*Dilara Sönmez* *Abdelrahman Elmughrabi*
Objectives: *This work package focuses on the design of the system architecture and AI pipeline. It includes defining the desktop application structure, model flow, and data pipeline, as well as preparing the SDD document. The aim is to translate the requirements into a workable design.*			
Tasks: ***Task 2.1 Architecture design:*** *Define the main components of the desktop application and their interactions.* ***Task 2.2 Model design:*** *Select and design the segmentation and classification approach.* ***Task 2.3 Data pipeline design:*** *Plan image preprocessing, input formatting, and output handling.* ***Task 2.4 SDD preparation:*** *Document the design decisions and system structure.*			
Deliverables ***D2.1:*** *System architecture design* ***D2.2:*** *Model design specification* ***D2.3:*** *System Design Document*			

Table 5. Work Package 3

WP 3: *Data Preparation and Preprocessing*
Start date: *28.03.2026* **End date:** *10.04.2026*

Leader:	*Dilara Sönmez*	**Members involved:**	*Dilara Sönmez* *Abdelrahman Elmughrabi*
Objectives: *This work package covers dataset collection, organization, preprocessing, and preparation for training and testing. The goal is to ensure that the mammography images are clean, consistent, and suitable for model development.*			
Tasks: ***Task 3.1 Dataset collection and license verification:*** *Collect the mammography datasets from public sources and verify their licenses and usage terms.* ***Task 3.2 Data cleaning and organization:*** *Clean the raw data and organize it into a consistent folder structure suitable for training and evaluation.* ***Task 3.3 Image preprocessing and format conversion:*** *Apply preprocessing operations such as resizing, normalization, and format conversion to prepare images for model input.* ***Task 3.4 Data split preparation:*** *Divide the dataset into training, validation, and test sets for model development and evaluation.*			
Deliverables ***D3.1:*** *Prepared dataset* ***D3.2:*** *Preprocessing pipeline* ***D3.3:*** *Finalized data split*			

Table 6. Work Package 4

WP 4: Segmentation and Classification Implementation			
Start date: 11.04.2026 **End date:** 24.04.2026			
Leader:	Abdelrahman Elmughrabi	**Members involved:**	*Dilara Sönmez* *Abdelrahman Elmughrabi*
Objectives: *This work package includes implementing the segmentation and classification models and improving their performance. The aim is to produce a working AI pipeline for suspicious region detection and benign/malignant classification.*			
Tasks: ***Task 4.1 Segmentation model implementation:*** *Implement and train the segmentation model for detecting and outlining suspicious regions in breast images. Task 4.2 Classification model implementation* ***Task 4.2 Classification model implementation:*** *Develop and train the classification model to determine whether the detected findings are benign or malignant.* ***Task 4.3 Model optimization and validation:*** *Tune model parameters, improve performance, and validate the models using appropriate evaluation metrics.* ***Task 4.4 Explainability integration:*** *Integrate methods to visualize the areas that affect the model's classification decisions.*			
Deliverables ***D4.1:*** *Implemented segmentation model* ***D4.2:*** *Implemented classification model* ***D4.3:*** *Trained model weights and validation results* ***D4.4:*** *Explainability output visualizations*			

Table 7. Work Package 5

WP 5: *Desktop Application Integration and Testing*			
Start date: *25.04.2026* **End date:** *22.05.2026*			
Leader:	*Dilara Sönmez*	**Members involved:**	*Dilara Sönmez* *Abdelrahman Elmughrabi*
Objectives: *The objective of this work package is to combine all previously developed components into a single functional desktop application. The preprocessing module, trained models, explainability component, and user interface will be integrated into one end-to-end workflow. After integration, the system will be tested from functional, integration, and usability perspectives. The main aim is to obtain a stable and demonstrable prototype before the supervisor demo.*			
Tasks: ***Task 5.1 Desktop application development:*** *Implement the desktop application interface that allows users to upload images and view segmentation, classification, and explanation results.* ***Task 5.2 Module integration:*** *Connect the preprocessing, inference, and output visualization modules into one end-to-end application flow.* ***Task 5.3 System testing and evaluation:*** *Perform functionality, integration, and usability testing to verify that the application operates correctly.* ***Task 5.4 Bug fixing and stabilization:*** *Resolve errors found during testing and improve the reliability and usability of the system before demonstration.*			
Deliverables ***D5.1:*** *Integrated desktop application prototype* ***D5.2:*** *End-to-end system workflow* ***D5.3:*** *Testing and evaluation results* ***D5.4:*** *Stable demo-ready version*			

Table 8. Work Package 6

WP 6: *Documentation, Thesis Writing, and Presentation*			
Start date: *09.05.2026* **End date:** *12.06.2026*			
Leader:	*Abdelrahman Elmughrabi*	**Members involved:**	*Dilara Sönmez* *Abdelrahman Elmughrabi*
Objectives: *The objective of this work package is to document the entire project and prepare all final course deliverables. This includes writing the thesis/report, organizing technical documentation, preparing the poster, and creating the final oral presentation materials. The results obtained from design, implementation, and testing will be clearly reported and discussed. By the end of this work package, the team will have a complete final submission package ready for evaluation and jury presentation.*			
Tasks: ***Task 6.1 Thesis writing:*** *Prepare the final report including introduction, requirements, design, implementation, testing, results, and conclusions.*			

Task 6.2 Technical documentation preparation: *Organize supporting documentation, figures, diagrams, screenshots, and final project records.* ***Task 6.3 Poster preparation:*** *Design the academic poster, summarizing the motivation, methodology, implementation, and outcomes of the project.* ***Task 6.4 Final presentation preparation:*** *Prepare presentation slides, demo flow, and speaking materials for the jury presentation.*
Deliverables ***D6.1:*** *Final thesis document* ***D6.2:*** *Complete technical documentation* ***D6.3:*** *Poster* ***D6.4:*** *Final presentation slides*

5. Glossary

Breast Imaging:
Medical imaging data of the breast used for detecting abnormalities such as masses, calcifications, or lesions.

Mammography:
A medical imaging technique that uses low-dose X-rays to examine breast tissue for early signs of cancer.

Lesion:
An abnormal area of tissue that may indicate disease or damage. In this project, lesions are suspicious regions in breast images.

Tumor:
An abnormal growth of tissue. A tumor may be benign or malignant.

Benign:
A non-cancerous abnormality that does not invade nearby tissues or spread to other parts of the body.

Malignant:
A cancerous abnormality that can invade surrounding tissues and may spread to other parts of the body.

Segmentation:
The process of identifying and outlining the exact region of interest in an image, such as a tumor or suspicious lesion.

Classification:
The task of assigning an input image or detected region to a category. In this project, classification is used to determine whether a finding is benign or malignant.

Region of Interest (ROI):
A selected part of an image that is considered important for analysis, such as a suspicious breast region.

Deep Learning:
A branch of machine learning based on neural networks with multiple layers, commonly used for image analysis and pattern recognition.

Convolutional Neural Network (CNN):
A type of deep learning model especially effective for image processing tasks such as detection, segmentation, and classification.

U-Net:
A deep learning architecture widely used in medical image segmentation because it can accurately localize structures in images.

Preprocessing:
The set of operations applied to raw data before model training or testing, such as resizing, normalization, contrast enhancement, and noise reduction.

Data Augmentation:
Artificially increasing the size and diversity of a dataset by applying transformations such as rotation, flipping, or scaling.

DICOM:
A standard medical imaging file format commonly used to store and transmit medical scan data and related metadata.

Explainability:
The ability of an AI system to provide understandable reasons or visual evidence for its predictions.

Grad-CAM:
A visualization technique used to highlight the image regions that most influenced a deep learning model's decision.

Dice Coefficient:
A performance metric commonly used in segmentation tasks to measure the overlap between the predicted region and the ground truth region.

Intersection over Union (IoU):
A metric used to evaluate segmentation performance by comparing the overlap between predicted and actual regions.

Sensitivity:
A classification metric that measures the ability of the system to correctly identify positive cases.

Specificity:
A classification metric that measures the ability of the system to correctly identify negative cases.

AUC (Area Under the Curve):
A performance metric used to evaluate how well a classification model distinguishes between classes.

Inference:
The process of using a trained model to make predictions on new, unseen data.

Desktop Application:
A software application that runs on a personal computer and provides the user interface for this project.

6. References

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